

ECON-7102

Environmental Economics I*

Prof. Casey J. Wichman
Georgia Institute of Technology

Fall 2026

Course materials: Canvas / course website (linked from Canvas)

Meeting time, classroom, and office hours: Check Canvas / course website

Course description: This is the first course in the Environmental Economics Ph.D. sequence. In this course, we will cover the theory of externalities and public goods related to the environment, followed by an exploration of policy instruments designed to address environmental problems. Next, we will revisit concepts of economic welfare and the tools economists use to value non-market environmental goods. The final portion of the course will survey relevant topics in contemporary environmental and energy economics—utility pricing, electricity markets, energy technologies, climate change, and conservation—to expose students to the research frontier.

*Prof. Casey Wichman: wichman@gatech.edu, School of Economics, Georgia Institute of Technology, 221 Bobby Dodd Way, Atlanta, GA 30332. **Last update:** August 24, 2026.

1 Course Overview

1.1 Course objectives

This course is intended to provide exposure to the theoretical foundations of environmental economics—particularly with respect to policy instrument choice and non-market valuation techniques—and give students a suite of tools to engage with the academic literature in environmental and resource economics. In addition, students will be exposed to the frontier of knowledge in the field and will develop a critical skillset to evaluate that research. Finally, students will take a step forward in identifying and pursuing a research question of their own interest that will, ideally, form the basis of a successful dissertation chapter or journal article.

1.2 Prerequisites

This course is an advanced Ph.D. course in economics intended to expose students to research at the frontier of the environmental and resource economics field. It is required that students have taken the first-year economics Ph.D. core courses in Microeconomic Theory (ECON 7012 & ECON 7013) and Econometrics (ECON 7022 & ECON 7023) (or equivalent). Students are expected to have a firm grasp of calculus and statistics.

1.3 Course goals

This course has three goals. First, students will learn fundamental concepts that underlie environmental and resource economics from a theoretical perspective. The first section of the course will cover the causes and consequences of environmental externalities and how to model potential solutions. Then, we will cover a variety of tools used to value environmental goods, with applications that include water and air quality. Second, students will be exposed to a variety of research at the frontier of the field, ranging from environmental justice to estimating climate change impacts, from consumer responses to electricity pricing to valuing health impacts of air pollution. Third, students will be expected to take a step towards producing, rather than consuming, research. In that effort, a subtext of this course will be in developing the craft of research, primarily focusing on how to identify and execute tangible research ideas. The second course in this sequence (ECON 7103 taught by Prof. Dylan Brewer) will focus more strongly on putting those ideas into action empirically.

2 Course Expectations

2.1 Attendance

Attendance is expected and will be necessary to succeed in the course. Participation in class discussions will be strongly encouraged. Ask questions. If you do not understand something in class, you are almost certainly not the only one. If you never volunteer an incorrect answer, you aren't taking enough risks.

Attendance is not graded directly. Poor performance in this dimension may affect the final grade.

2.2 Course website and Canvas

The course schedule, slides, and journal-article readings live on the course website, which is linked from the Canvas home page. The website is the current version of the schedule and reading list and will be updated as the semester goes on.

Canvas is used for assignment submissions, grades, announcements, and reference materials such as textbook PDFs.

2.3 Grading

The final grade is calculated as follows:

"Weekly" reading responses	20 pts
Research paper proposal	40 pts
Midterm	25 pts
Lead in-class discussion	5 pts
Referee report on JMP	10 pts
Total	100 pts

Assignments are described in Section 3.

There is a degree of subjectivity in grading graduate coursework. I expect that all students who put an appropriate amount of effort into this course will receive an A or B. In the event that a student demonstrates a severe lack of effort, a lower grade may be appropriate.

2.4 Required course materials

The primary required text is Phaneuf & Requate (2017), *A Course in Environmental Economics: Theory, Policy, and Practice*, Cambridge University Press (ISBN 9780511843839, <https://doi.org/10.1017/9780511843839>). All other readings will be provided electronically, on the course website or on Canvas. There are no other required purchases.

2.5 Additional criteria for successful completion

Students must complete all graded components (reading responses, research proposal, midterm, discussion lead, and referee report) to pass the course. Late assignments will not receive credit without a valid excuse.

2.6 Academic Integrity

Cheating is unacceptable. You are hereby reminded that you have pledged to uphold the honor code as follows:

Having read the Georgia Institute of Technology Academic Honor code, I understand and accept my responsibility as a member of the Georgia Tech community to uphold the Honor Code at all times. In addition, I understand my options for reporting honor violations as detailed in the code.

Should you be caught cheating in this class you will be prosecuted according to the honor code and policies and procedures established by the Honor Advisory Council. Should you have any questions about this do not hesitate to contact me.

2.7 Student-Faculty Expectations Agreement

At Georgia Tech, it is important to strive for an atmosphere of mutual respect, acknowledgment, and responsibility between faculty members and the student body. See <http://www.catalog.gatech.edu/rules/22/> for an articulation of some basic expectation that you can have of me and that I have of you. In the end, simple respect for knowledge, hard work, and cordial interactions will help build the environment we seek. Therefore, I encourage you to remain committed to the ideals of Georgia Tech while in this class.

2.8 Accommodations for students with disabilities

If you are a student with learning needs that require special accommodation, contact the Office of Disability Services at (404) 894-2563 or <http://disabilityservices.gatech.edu/>, as soon as possible, to make an appointment to discuss your special needs and to obtain an accommodations letter. Please also e-mail me as soon as possible to set up a time to discuss your learning needs.

3 Details on course assignments

“Weekly” reading responses (20 points)

Students write a short response paper, roughly 250–300 words or two paragraphs, on designated readings. These readings are marked **[response due]** on the course website schedule (linked from Canvas). Expect eight to ten over the semester.

The response needs to go beyond summarizing the reading. I’ve already read the article and I do not need it summarized. Make connections to other topics in the course, to stories in the news, or to ongoing policy discussions. Reflect on how the reading updated your thinking. If you disagreed with a conclusion or a fundamental assumption, expand on that.

Focus on three questions:

1. What was one thing you enjoyed about the article?
2. What was one thing you did not understand, or would like to discuss in class?
3. What extensions or applications of the article would be interesting?

Submit via Canvas by 11:59 PM ET the day before the class in which the reading is assigned. That timing lets me read the responses before class. Late responses receive half credit.

Research paper proposal (40 points)

One goal of this course is to provide the tools necessary to conduct your own research in environmental, energy, and resource economics. The proposal is built in two stages. In the first stage, students generate and pitch several research ideas. In the second stage, students develop the most promising idea into a full proposal.

Stage 1: Research abstracts (five, 2 points each). A one-page abstract is due every two weeks (on Sundays): September 13, September 27, October 11, October 25, and November 8, by 11:59 PM ET. Each abstract uses a fixed template:

Research abstract template

1. Question & motivation, stated plainly and clearly
2. Source of variation, or the key mechanism if the project is theoretical
3. Predicted hypotheses/findings
4. Data needed, and whether the data exist or are accessible
5. Closest existing paper and how this paper differs
6. Kill criterion: what are the major hurdles that would make this project infeasible?

The sixth item is important. I want you to learn to “fail quickly.” Most research ideas fail, and there is value in developing failure as a skill. The purpose of this exercise is to make failure fast and cheap, before you have spent a year (or more) on a question that was never going to work.

Each abstract submission may present entirely new ideas or variations on an earlier one, with the changes clearly marked. That said, generating five distinct ideas is encouraged.

Rapid-fire pitches. In the class following each abstract deadline, every student pitches their idea in one to two minutes. The point is to put your idea in front of the room and, ideally, to generate side conversations after class.

Final proposal pitch (Monday, November 16, in class, 5 points). Each student gives a five-minute pitch on the idea they intend to develop, followed by discussion.

Stage 2: Full proposal. Develop the chosen idea into a complete proposal. The written proposal is due Friday, December 11. The proposal does not need data or empirical results, but should include:

Proposal template

1. An introduction that clearly states the research question and anticipated contributions to existing literatures.
2. An economic framework or model that motivates and crystallizes the question.
3. A data section, if applicable, describing potential data sets.
4. An empirical strategy section, if applicable, describing empirical hypotheses and identification frameworks.
5. A discussion of the limitations of the approach, for example data access, lack of a causal identification strategy, or null results.

The final proposal should “look like” the front half of a journal article or working paper. It should be written in a style that is accessible to economists outside your subfield. The proposal should be 10–15 pages, double-spaced, with 12-point font and 1-inch margins.

In the last few classes of the semester, students are expected to present their final proposals to the class in a 15-20 minute presentation, followed by discussion.

Milestone	Due	Points
Research abstracts (five)	Biweekly Sundays, Sep. 13 – Nov. 8	10
Final proposal pitch	November 16, in class	5
Presentation	Last three classes	5
Written proposal	December 11	20

Collaboration. Every graded item in this course is submitted individually. But, you should be collaborating with your peers. Read each other’s abstracts, talk about them outside class, and offer help when you know a paper or a dataset a classmate needs. Start jointly developing research ideas with your classmates. This is how economics research works in practice. Look at any recent issue of a major economics journal and count the number of solo-authored papers.

Use of AI. Do not use AI tools to generate your research questions or to supply your motivation. LLMs are better than you at programming, developing theory, and (probably) writing. We still have a comparative advantage in creativity. If you outsource your idea generation to an LLM, think about what that means for the value of your Ph.D. Do use LLMs to interrogate your ideas. Ask for the ten closest papers to your idea and verify every one. Ask for the three strongest reasons your paper should be rejected. Ask which assumption in your model is “load-bearing.” Ask which data sets might be useful. Use LLMs to help you think critically about your work. Use AI as a complement, not a substitute, for your own thinking.

Students should not use this proposal in conjunction with a writing assignment for another course. Proposals can sit at the intersection with other fields, for example environment and health, environment and development, or environment and trade.

Midterm (25 points)

In class, Class 15 (Monday, October 19).

Lead in-class discussion (5 points)

Each student leads or co-leads discussion of one paper in the topics section of the course.

Referee report on a job-market paper (10 points)

Students find a job-market paper by a current Ph.D. candidate working on an environmental, energy, or resource economics topic, and write a referee report. The goal is to see the end product of the Ph.D. on a topic of interest and to practice giving critical but constructive feedback.

I will provide guidance during the semester on where to search for job-market papers and on the structure of a referee report.

Course schedule and readings

The class-by-class schedule—dates, topics, deadlines, slides, and the reading list—lives on the course website, linked from Canvas. The website is always the current version. The reading list is organized by class meeting, with PDFs. The list is tentative and will change with the pacing of the course. Each week, I will communicate where to focus your reading effort, for example on particular sections within chapters or papers that are assigned.